

PUMPING MECHANISMS, VALVE SYSTEMS, AND FLUID CONTROL ARCHITECTURES

In various embodiments, the intraluminal compliance and distance profiling device incorporates fluid control architectures configured to inflate, deflate, regulate, or stabilize expandable regions used for compliance measurement. Pumping mechanisms may be implemented using manual, mechanical, electromechanical, or hybrid approaches while maintaining functional equivalence.

Manual pumping embodiments may include squeeze bulbs, hand-actuated reservoirs, syringe-style displacement mechanisms, or elastic recovery chambers configured to introduce controlled air or fluid volume into inflatable regions. Manual systems may provide low-cost operation while allowing tactile user control over expansion.

In certain embodiments, automated pumping mechanisms may include diaphragm pumps, micro-compressors, piezoelectric pumps, electrostatic micro-pumps, or MEMS-based displacement systems. Automated pumps may enable programmable inflation sequences, closed-loop pressure regulation, or remote software control of expansion behavior.

Valve systems may include check valves, proportional valves, electronically actuated valves, pressure relief valves, or passive flow restrictors configured to regulate fluid movement. Valves may prevent backflow, maintain stable pressure during measurement hold phases, or enable rapid emergency deflation.

Fluid control architectures may incorporate staged inflation pathways allowing incremental expansion across multiple chambers or regions. Multi-chamber configurations may distribute expansion forces more uniformly and may allow localized compliance measurements along different portions of the insertable structure.

In certain implementations, pressure equalization channels may stabilize internal pressure following pump actuation. Equalization systems may reduce oscillation, dampen pressure spikes, and improve repeatability of compliance measurements.

The device may further include passive safety elements such as rupture disks, compliant expansion limits, mechanical stops, or pressure-limited regulators preventing over-expansion independent of electronic control systems.

In some embodiments, fluid control components are modular and replaceable, allowing service, upgrade, or adaptation to alternative pumping technologies. Modular fluid architectures enable migration between manual consumer configurations and automated research or clinical-support configurations.

Control algorithms executed by onboard electronics or associated software may coordinate pump actuation, valve timing, inflation rate control, and deflation sequencing. Algorithmic control may optimize measurement stability, user comfort, and safety simultaneously.

Accordingly, pumping mechanism and valve system embodiments provide comprehensive support for fluid regulation strategies enabling safe, repeatable expansion behavior while preventing design-around implementations based solely on alternative pump or valve technologies.